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Experimental investigation on the vertical vortex-induced vibration of a long-span continuous steel box girder bridge

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ABSTRACT

This study provides a comprehensive investigation into the vertical vortex-induced vibration (VIV) of a long-span continuous steel box girder bridge under skew wind conditions, emphasizing critical insights for real-world wind scenarios. Wind tunnel tests on 1:30 sectional and full-bridge aeroelastic models are conducted to explore VIV behavior under varied turbulence intensities (I_u), angles of attack (α_m), Scruton numbers (Sc), and skew wind angles (β). The tests systematically examine lock-in wind regions, vibration amplitudes, and frequencies, providing refined data essential for resilient bridge design. The findings highlight the limitations of conventional Independence Principle (IP) theory, which fails to predict VIV responses under skewed winds ($\beta \neq 0^\circ$). Results reveal that as β increases, VIV amplitude reduces and onset wind speed shifts, highlighting skew wind angles's mitigating effect on VIV. VIV is most intense at $\beta = 0^\circ$, underscoring the need for alternative predictive methods beyond IP theory in skew wind conditions. Increasing α_m intensifies the lock-in region and delays cut-off speeds, validating the aspect of VIV behavior. The maximum amplitude is highly notable when $I_u < 5.67\,\%$, and it only slightly decreases compared to the smooth flow case. A 73 % amplitude reduction is achieved as damping increases (Sc from 0.53 to 32.84), offering a quantitative benchmark for bridge design resilience. In the full-bridge model, VIV ceases at $Sc \geq 3.344$, establishing a critical threshold for eliminating VIV in practical designs.

1. Introduction

Bridge vortex-induced vibration (VIV) is a type of limit cycle oscillation, combining forced and self-excited dynamics. Due to its unique structural characteristics, long-span steel box girder bridges, characterized by their light weight, flexibility, blunt geometry, and low damping, are especially susceptible to significant vertical VIV under moderate-to-high wind speeds during both construction and operational stages. For example, at wind speeds of 11.6 ~ 15.6 m/s, the Volga River Bridge in Russia experienced large VIVs with a maximum amplitude of 40 cm and a frequency range of 0.45 ~ 0.68 Hz [1]. The Tokyo-Bay sea-crossing Bridge in Japan experienced vertical VIVs with a maximum amplitude of 50 cm at wind speeds of 14 ~ 17 m/s [2]. Vertical VIVs with a maximum amplitude of 10 cm occurred at wind speeds of 32 ~35 m/s on the Hong Kong–Zhuhai–Macao Bridge's deep-water non-navigable section in China [3]. The Xihoumen Bridge has frequently experienced vertical VIV at wind speeds of 6 ~ 10 m/s. The reported maximum vibration amplitude is over 20 cm [4–6]. Although VIV does not directly lead to catastrophic failure of the structure, it may affect driving comfort and safety, cause fatigue damage to structural components and lead to extensive adverse social effects [7,8]. As a result, the vertical VIVs of long-span steel box girder bridges have attracted increasing attention from researchers. In addition, VIV control is also an important area of research, and VIV control countermeasures can be divided into three categories: aerodynamic, structural, and mechanical countermeasures [9,10]. Notably, no torsional VIV cases have been reported for this type of bridge, and therefore, torsional VIV is not considered in this study.

Most existing studies focus on VIV under idealized perpendicular winds, leaving a gap in understanding VIV under skewed winds. Traditionally, skew winds were assumed to reduce VIV responses. Nevertheless, recent wind tunnel and flume studies on cylindrical models reveal that VIV responses can peak at non-perpendicular ($\beta \neq 0^\circ$) wind angles [11–13]. This possibility highlights the need to verify if similar behavior is present in bridge cross-sections, especially given the complex real-world wind conditions that are still not fully understood. Skew wind conditions may influence VIV dynamics in ways not covered by traditional assumptions, requiring precise studies to confirm or

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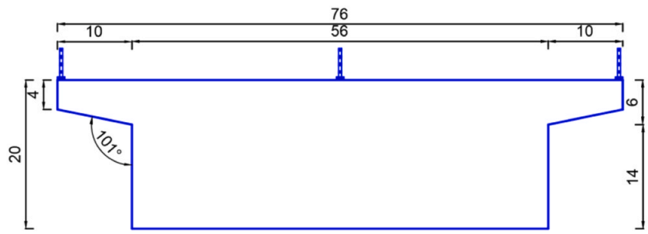

Fig. 1. Geometry of the sectional model (Unit: cm).

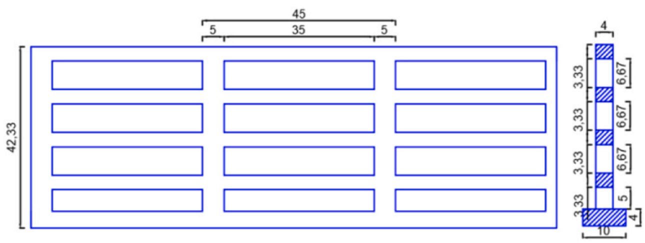

Fig. 2. Geometry of the railing (Unit: mm).

Fig. 3. The sectional model.

Table 1
Parameters of the sectional model.

Parameter	Symbol	Value	Unit
Length	L	3.94	m
Width	B	0.76	m
Height	D	0.2	m
Equivalent mass	m_1	13.815	kg/m
Natural frequency	f_s	3.158	Hz
Mechanical damping ratio	ξ_1	0.055 %	1
Air density	ρ	1.225	kg/m^3

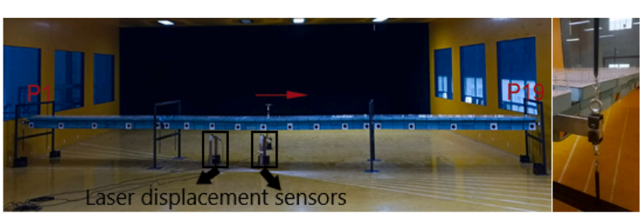

Fig. 4. Full-bridge aeroelastic model and hoisting method.

modify existing theoretical models.

Wind tunnel testing, including sectional model tests and full-bridge aeroelastic model tests, is currently the most widely used approach for studying bridge VIV. Sectional model testing is favored for its simplicity and cost-effectiveness, often incorporating measurements of vibration, pressure, velocity, and force [6,14–16]. Full-bridge aeroelastic model tests can simulate the bridge's multi-order modes, providing a more accurate depiction of the structure's aerodynamic stability and wind-induced vibration response in real atmospheric boundary layers [17]. Combining these two methods allows a comprehensive view of bridge VIV, incorporating both structural and aerodynamic effects.

Turbulence intensity (I_u) and mechanical damping (ξ) are essential in controlling VIV. High I_u disrupts vortex coherence, reducing VIV response, while ξ affects both lock-in regions and VIV amplitudes [18–21]. Most bridge VIV studies assume that the critical wind direction is perpendicular to the bridge axis, yet in natural settings, the yaw angle (β, which represents the angle between the direction perpendicular to the model's central axis and the wind's direction) ranges from 0° to 360° [22]. Van Atta [23] proposed the independence principle (IP) to evaluate VIV responses when $\beta \neq 0°$, assuming that the flow component parallel to the structure axis is negligible. This theory is valid at small yaw angles, but notable experimental deviations are significant for β angles around 45° to 60° [24]. Numerous studies on cylinders with yaw angles in wind tunnels or flumes [11–13] have shown that the maximum VIV response of a circular cylinder occurs when $\beta \neq 0°$ [13]. Wang et al. [22] comprehensively investigated VIV responses of a 4:1 rectangular cylinder across various yaw angles (β) and angles of attack (α_m), concluding that conventional IP theory is ineffective for skew winds, emphasizing the need for bridge section testing to validate IP under diverse yaw angles.

This study fills gaps in understanding bridge VIV under skew winds through comprehensive wind tunnel testing on a long-span continuous steel box girder bridge. Key parameters—including wind yaw angles (β), wind angles of attack (α_m), turbulence intensity (I_u), and mechanical damping (ξ)—are systematically varied. This approach uniquely integrates the effects of α_m, I_u, and β into the VIV analysis framework, providing a comprehensive and realistic evaluation of VIV behavior under skew winds. This multi-dimensional analysis provides refined theoretical insights and critical data for engineers to design structures that withstand complex, real-world wind scenarios.This study is one of the few to systematically investigate VIV under skew winds, providing evidence that skew wind conditions do not increase VIV responses. The findings establish a benchmark for future studies to accurately calibrate models and explore complex wind-structure interactions. This benchmark is crucial for designing robust, aerodynamically stable bridges that withstand complex real-world wind patterns.

This work thoroughly investigates the VIV responses of a long-span continuous bridge with a steel box girder. The investigation uses wind tunnel experiments with the 1:30 scale sectional and full-bridge aeroelastic models. The influences of β, α_m, I_u, and ξ on the VIV responses are investigated. Section 2 introduces the test setup, cases, and process. Section 3 provides a systematic presentation of the VIV characteristics of both the sectional and full-bridge aeroelastic models, considering different cases. An analysis is carried out on the VIV lock-in region, amplitudes, frequencies, and the suitability of IP when $\beta \neq 0°$. Finally, the main findings from this study are summarized in Section 4.

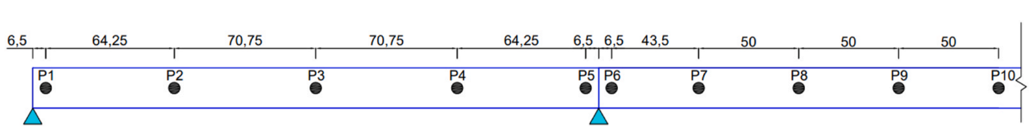

Fig. 5. Measuring point arrangement (Unit: cm).

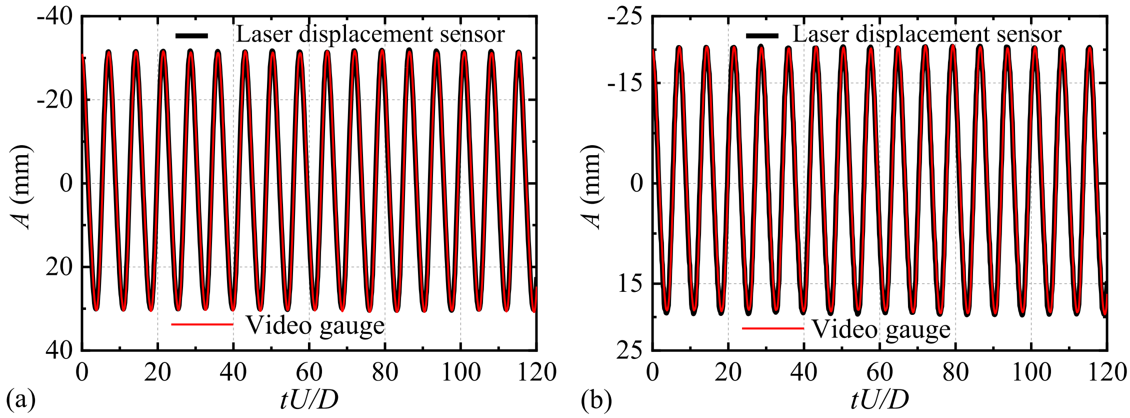

Fig. 6. Displacement histories:(a) P8; (b) P10.

Table 2
Comparison of displacement measured by different acquisition instruments.

Equipment	Amplitude		Standard deviation	
	P8	P10	P8	P10
video gauge	19.92	28.62	14.05	21.89
laser displacement sensors	19.81	28.87	14.12	21.72

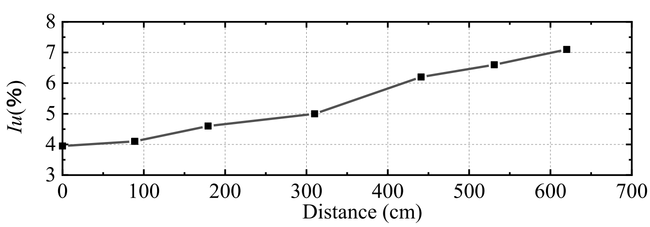

Fig. 7. The turbulence intensities at different points along the bridge deck when β= 40°.

Table 3
Testing cases for sectional model.

Flow field	ξ (%)	$Sc = 4\pi M\xi_1/(\rho BDL)$	α_m	Case
Smooth	0.055	0.53	−5°	S-A1
			−3°	S-A2
			0°	S-A3
			+ 3°	S-A4
			+ 5°	S-A5
Turbulence I_u= 4.27 %	0.055	0.53	−5°	S-B1
			−3°	S-B2
			0°	S-B3
			+ 3°	S-B4
			+ 5°	S-B5
Turbulence I_u= 5.67 %	0.055	0.53	−5°	S-C1
			−3°	S-C2
			0°	S-C3
			+ 3°	S-C4
			+ 5°	S-C5
Smooth	0.056	0.54	0°	S-D1
	0.066	0.63	0°	S-D2
	0.422	4.05	0°	S-D3
	0.485	4.65	0°	S-D4
	0.772	7.41	0°	S-D5
	1.221	11.72	0°	S-D6
	1.732	16.63	0°	S-D7
	1.835	17.62	0°	S-D8
	3.421	32.84	0°	S-D9

Table 4
Testing cases for full-bridge aeroelastic model.

Flow field	ξ (%)	$Sc = 4\pi M\xi_2/(\rho BDL)$	β	α_m	Case
Smooth	0.67	0.604	0°	−5°	FA−0–1
				−3°	FA−0–2
				0°	FA−0–3
				+ 3°	FA−0–4
				+ 5°	FA−0–5
			10°	−5°	FA−10–1
				0°	FA−10–2
				+ 5°	FA−10–3
			20°	−5°	FA−20–1
				0°	FA−20–2
				+ 5°	FA−20–3
			30°	−5°	FA−30–1
				0°	FA−30–2
				+ 5°	FA−30–3
			40°	−5°	FA−40–1
				0°	FA−40–2
				+ 5°	FA−40–3
Turbulence	0.67	0.604	0°	−5°	FB−0–1
				−3°	FB−0–2
				0°	FB−0–3
				+ 3°	FB−0–4
				+ 5°	FB−0–5
			10°	−5°	FB−10–1
				0°	FB−10–2
				+ 5°	FB−10–3
			20°	−5°	FB−20–1
				0°	FB−20–2
				+ 5°	FB−20–3
			30°	−5°	FB−30–1
				0°	FB−30–2
				+ 5°	FB−30–3
			40°	−5°	FB−40–1
				0°	FB−40–2
				+ 5°	FB−40–3
Smooth	2.23	2.104	0°	−5°	FC−0–1
				−3°	FC−0–2
				0°	FC−0–3
				+ 3°	FC−0–4
				+ 5°	FC−0–5
Smooth	3.71	3.344	0°	0°	FD−0–1
				+ 3°	FD−0–2
				+ 5°	FD−0–3

2. Experimental setup

The experiments were conducted in the boundary layer wind tunnel laboratory at the Changsha University of Science and Technology in China. The sectional model tests were conducted in the high-speed test section (the length, width and height were 21 m, 4 m and 3 m, respectively). The full-bridge aeroelastic model tests were conducted in the

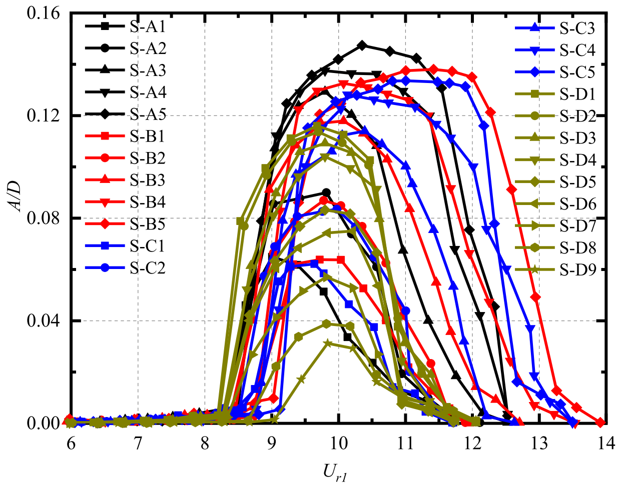

Fig. 8. VIV amplitudes of the sectional model.

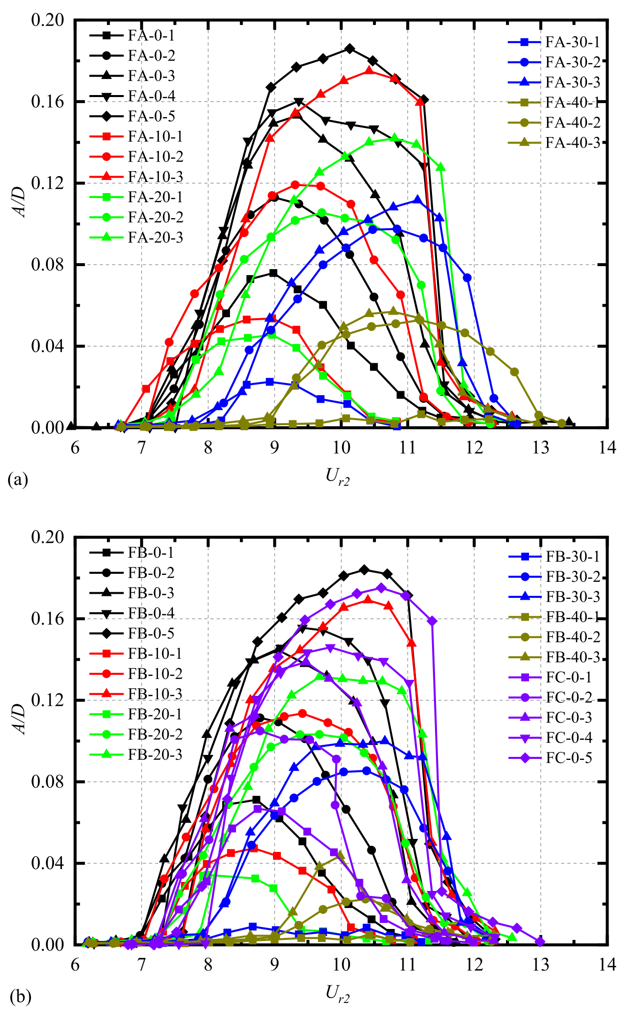

Fig. 9. VIV amplitudes of full-bridge aeroelastic model: (a) Case group FA; (b) Case group FB and FC.

low-speed test section (the length, width and height were 21 m, 10 m and 3 m, respectively). The smooth flow field generated in the wind tunnel can meet the test requirement that the I_u of the wind field is less than 0.5 %.

2.1. Sectional model and measuring instruments

Fig. 1 displays the geometric configuration of the sectional model (with a height (D) of 200 mm, a width (B) of 760 mm, and a length (L) of 3940 mm). Fig. 2 displays the intricate composition of the railing. An aluminum core beam guarantees the stiffness of the sectional model, while a high-density foam board and acrylic plate imitate the aerodynamic design. The railings are carved from acrylic plates. The vibrating portion has an equivalent mass of m_1= 13.815 kg/m. The sectional model test achieves a maximum blocking ratio of 8.9 % when $\alpha_m = \pm 5^\circ$. The blockage ratio is relatively large but it is difficult to solve this issue due to the limitation of the wind tunnel size. The blockage ratio of the sectional model is generally not higher than 5 %. To meet this requirement, the model scaling ratio will be too small, and hence not only the detailed structure of the model cannot be strictly simulated, but also the Reynolds number effect is obvious. The wind speeds at the model are also corrected by the blocking ratio. External damping in the sectional model test is introduced by wrapping a double-sided adhesive layer around the suspension springs. The damping level is adjusted by varying the number of wraps and the thickness of the adhesive layer. This method has been tested, and the resulting damping is found to be linear.

The sectional model was supported by eight springs (Fig. 3), collectively possessing a rigidity of 21430 N/m. Two ends of the sectional mode are very close to the wind tunnel walls and hence the wind tunnel walls work as the end plates. During the sectional model testing, a grille was constructed at the wind tunnel outlet to generate the turbulence field. The turbulence intensity was controlled by adjusting the size and number of the horizontal and vertical beams in the grille. A 3D pulsating anemometer was used to measure the turbulence intensity at multiple wind speeds, ensuring the accuracy of the experiments. The displacements of the sectional model were measured using three laser displacement sensors. The three sensors were all positioned beneath the model. Laser displacement sensors A and B were symmetrically arranged on one side of the model, 40 cm apart and 30 cm away from the model's end. Sensor C is symmetrically positioned with respect to sensor A about the center of the sectional model. The laser displacement sensors (optoNCDT1750) have a measurement range of 50 mm, a dynamic precision of 0.01 mm, and a sampling frequency of 200 Hz.

Using incremental steps of 0.2 m/s, the wind speed was changed from 2 to 10 m/s during the test. The dynamic viscosity (ν) and the inflow wind velocity (U) determine the Reynolds number (Re=UD/ν), which varies between 2.14×10^4 and 1.07×10^5. For all cases, the sectional model's displacements are captured for more than 30 seconds at every wind speed once the vibration stabilizes. Table 1 summarizes the primary geometric and dynamic features of the sectional model.

2.2. Full-bridge aeroelastic model and measuring instruments

The span arrangement of the full-bridge aeroelastic model is 283 cm + 400 cm + 283 cm. The geometry of the bridge deck and the detailed structure of the railing for the full-bridge aeroelastic model are shown in Figs. 1 and 2. An aluminum core beam provides the stiffness for the model, while high-density foam and acrylic plates replicate the aerodynamic design. The railings are carved from acrylic plates, and the physical mass of the model is 5.975 kg/m. The standard segment is 20 cm long, and the non-standard segments, either 16.2 cm or 20.4 cm in length, are positioned at the ends of each span. Two aerodynamic coat segments are spaced 2 mm apart to prevent collisions during vibrations induced by the wind. In this paper, the first-order vertical bending VIV of the model is examined. Therefore, a vertical two-way suspension method is proposed to replace the conventional support system, and 16 connecting rods are used to hoist the model, as shown in Fig. 4. The turbulence field simulation in the full-bridge aeroelastic model experiments is consistent with that used in the sectional model experiments. After modeling and verification, the above hoisting method will not

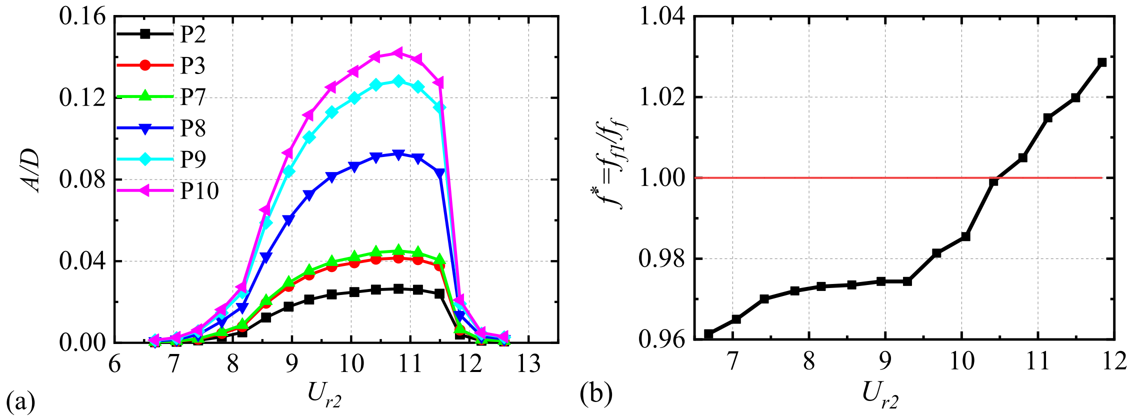

Fig. 10. The VIV responses for FA-20–3 case: (a) Dimensionless amplitudes; (b) The evolution of f^*.

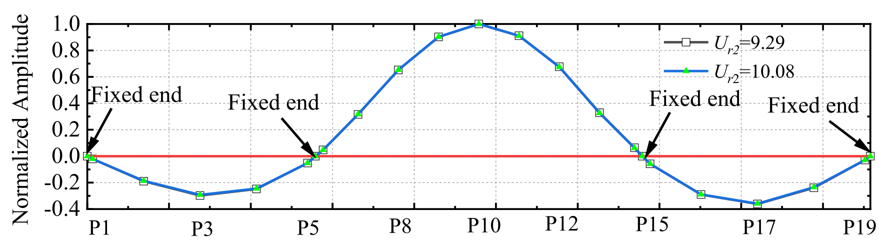

Fig. 11. Mode shape constructed based on vibration data at available measuring sections.

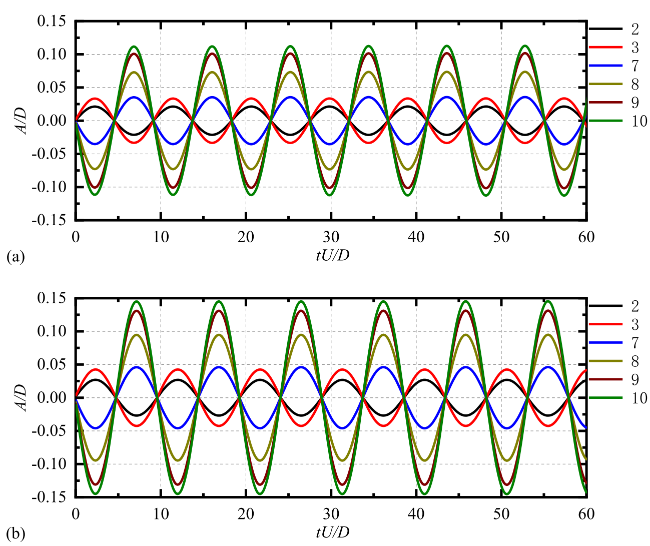

Fig. 12. Displacement time histories: (a) $U_{r2} = 9.29$; (b) $U_{r2} = 10.08$.

substantially impact the first-order mode and formation of a long-span continuous girder bridge.

The full-bridge aeroelastic model's time-varying motions are captured using a video gauge (FLIR ORX-10G-51S5M-C) with a sample frequency of 200 Hz. The model has 19 measurement sections (P1 to P19) arranged on it, and the P1-P10 measuring points are arranged as shown in Fig. 5. The P11-P19 measuring points and the P1-P10 measuring points are symmetrical about the P10 measuring points. The advantage of the video gauge over the laser displacement sensor is that it can reduce the interference of the flow field. Two laser displacement sensors are also used to measure the displacements at 1/4 and 1/2 spans of the second span to confirm the correctness of the measurements made by the video gauge. The first-order symmetrical vertical mode's natural frequency, modal mass, and damping ratio are determined as f_f= 3.43 Hz, M_e= 1.298 kg/m, ξ= 0.67 %, and Re range is (2.14×10^4, 1.07×10^5), respectively. The sectional model test achieves a maximum blocking ratio of 8.9 % when $\alpha_m = \pm 5^\circ$. In the full-bridge aeroelastic model test, external damping is introduced by applying transparent tape to the bridge deck. The damping level is adjusted by modifying the taped area and thickness, and the resulting damping has been tested and found to be linear.

Under the condition of smooth flow, $\beta = 0^\circ$, $\alpha_m = 0^\circ$, $U = 4.91$ m/s, and $\xi = 0.67$ %, the displacement histories of P8 and P10 measured by video gauge and laser displacement sensors are compared in Fig. 6, and the amplitudes and standard deviations are shown in Table 2. From Fig. 6 and Table 2, the statistical values and phases of the two sets of displacement histories are consistent. The amplitudes and standard deviations are very close, suggesting that the video gauge can measure the displacement of the full-bridge aeroelastic model with high accuracy.

2.3. Testing cases

The 1:30 sectional and full-bridge aeroelastic models are used to study the VIV characteristics of the bridge deck under different ξ, α_m, β, and I_u. If the sectional model is rotated in a clockwise direction, the value of α_m is positive. In the full-bridge aeroelastic test, the I_u at different points on the bridge deck are different if β is non-zero. As an example, the I_u at various points along the bridge deck is in Fig. 7 for the case of $\beta = 40^\circ$ (The leading edge of the bridge section away from the wind outlet is 0 point when $\beta = 40^\circ$, and the center position of the model at other β is consistent with that at $\beta = 40^\circ$, so the I_u at other β can be calculated according to $\beta = 40^\circ$).

The testing cases of the sectional and full-bridge aeroelastic models are listed in Tables 3 and 4, respectively. The sectional and full-bridge model tests are divided into 4 groups according to the test settings. In the case name, the first letter (S or F) respectively indicates the sectional model or a full-bridge aeroelastic model. The S-A, S-B, and S-C groups are test studies of different flow fields with different α_m under $S_C = 0.53$, and the S-D group is test studies of different S_C at $\alpha_m = 0^\circ$ under smooth flow. The FA and FB groups are test studies of different flow fields with

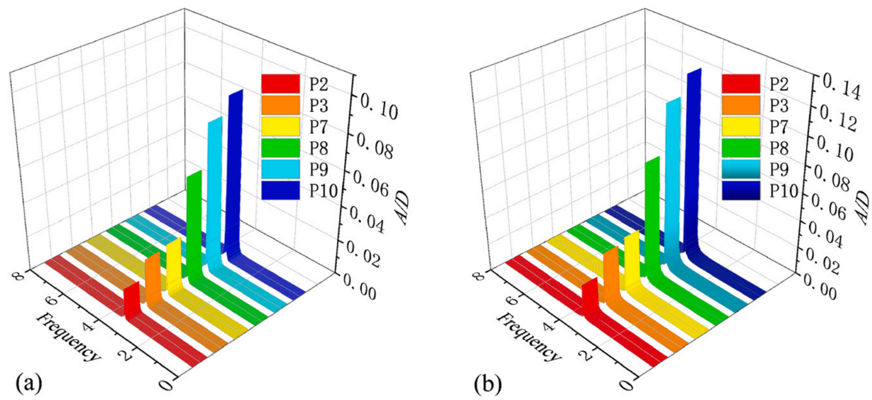

Fig. 13. Amplitude spectra of VIV: (a) $U_{r2} = 9.29$; (b) $U_{r2} = 10.08$;.

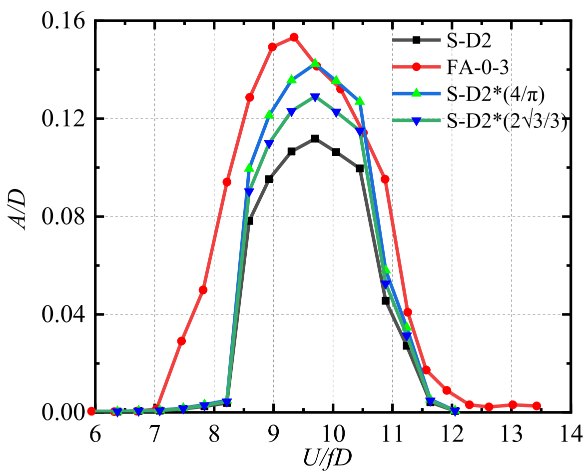

Fig. 14. The VIV responses under similar S_C numbers.

different α_m and β under $S_C = 0.604$. The FC and FD groups are test studies of different α_m under smooth flow, and the S_C numbers are 2.104 and 3.344, respectively. The S-A, S-B and S-C groups are designed to study the influences of I_u and α_m on the VIV responses. The S-D group is designed to study the influence of Sc on the VIV responses. The FA and FB groups are designed to study the influences of I_u, β, and α_m on the VIV responses. The FC and FD groups are designed to study the influence of ξ on the VIV responses.

3. Test results and discussions

3.1. VIV characteristics

Fig. 8 displays the relationship between the dimensionless VIV amplitude A/D (where A represents vertical vibration amplitude) and the reduced wind velocity $U_{r1}=U/f_sD$ (where U is the mean wind velocity) for the different cases of the sectional model. Fig. 9 displays the VIV amplitudes versus the reduced wind velocity $U_{r2}=U/f_fD$ for all testing cases measured at the P10 section of the full-bridge aeroelastic model. For the full-bridge aeroelastic model, the focus is primarily on the maximum VIV amplitude. Unless otherwise specified, the vibration amplitudes in the following sections are based on the data from the P10 measurement point. The I_u, β, α_m, and ξ (or equivalently, the Sc number) significantly impact the A/D and lock-in region.

The VIV responses (FA-20–3) at P2, P3, and P7 to P10 are shown in Fig. 10(a), and Fig. 10(b) displays the dimensionless vibration frequencies ($f^* = f_{f1}/f_f$) plotted against the reduced wind speed.

In Fig. 10(a), the amplitude curve form, cut-off wind speed, and onset wind speed of each measuring section in Fig. 10(a) are all consistent under the same U_{r2}. In Fig. 10(b), the f^* increases slightly as the U_{r2} increases. The mode shape constructed based on the vibration data at available measuring sections when $U_{r2} = 9.29$ and $U_{r2} = 10.80$ are shown in Fig. 11. At different U_{r2}, the mode shape of the full-bridge aeroelastic model remains consistent, and the modal shape exhibits symmetrical about the middle span. Fig. 12 and Fig. 13 are the time histories and amplitude spectra of the VIV displacement responses of P2, P3, P7, P8, P9, and P10 at $U_{r2} = 9.29$ and $U_{r2} = 10.80$, respectively. Regarding the symmetrical monitoring points, their responses in terms of amplitude and phase are indeed similar. As for points P1, P5, and P6, the response patterns are consistent with those of the other points. However, since these points are located near the fixed end, their vibration amplitudes are relatively small and therefore not prominently displayed in the figure.

As shown in Figs. 12 and 13, the vibration of each measuring point is very stable and is a mono-frequency vibration under skew winds, which conforms to the self-limiting and lock-in characteristics of VIV. The sectional model test is usually used to predict and evaluate the VIV performance of the full bridge. However, the full bridge is a three-dimensional structure, and its VIV performance is affected by the vibration mode and the spanwise correlation of the vortex-induced force. Sato [25] and Zhitian et al. [26] proposed using an amplitude correction coefficient to calculate the VIV amplitude of the full bridge. Assuming that the vortex-induced force is completely correlated along the spanwise direction, the amplitude amplification coefficient is $4/\pi$ (derived from a linear vortex vibration force model) or $2\sqrt{3}/3$ (derived from a nonlinear vortex vibration force model). The S_C numbers of the FA-0–3 and S-D2 cases are 0.604 and 0.63, respectively. Thus, the sectional model amplitude of case SD-2 is used to predict the full-bridge VIV amplitude of case FA-0–3 (P10 measuring point) based on the linear and nonlinear amplitude amplification coefficients, as shown in Fig. 14.

In Fig. 14, the lock-in region of the sectional model is enveloped in the lock-in region of the full-bridge aeroelastic model. The cut-off wind speed is basically the same, but the onset wind speed is more delayed than the full-bridge aeroelastic model. The amplitude of the sectional model corrected by the linear or nonlinear amplitude correction coefficient is still smaller than that of the full-bridge aeroelastic model, but the difference is insignificant. The differences may be ascribed to various experimental uncertainties, e.g., the model fabrication accuracy, geometric dimension, vibration modes, structural dynamics, spanwise correlation, boundaries, installation status, flow conditions (e.g., angle of attack, yaw angle, turbulence intensity, wind velocity), and apparatus measurement accuracy.

3.2. Influence of β on VIV and the applicability of IP theory

The influences of wind yaw angle ($\beta = 0^\circ$, 10°, 20°, 30°, and 40°) on

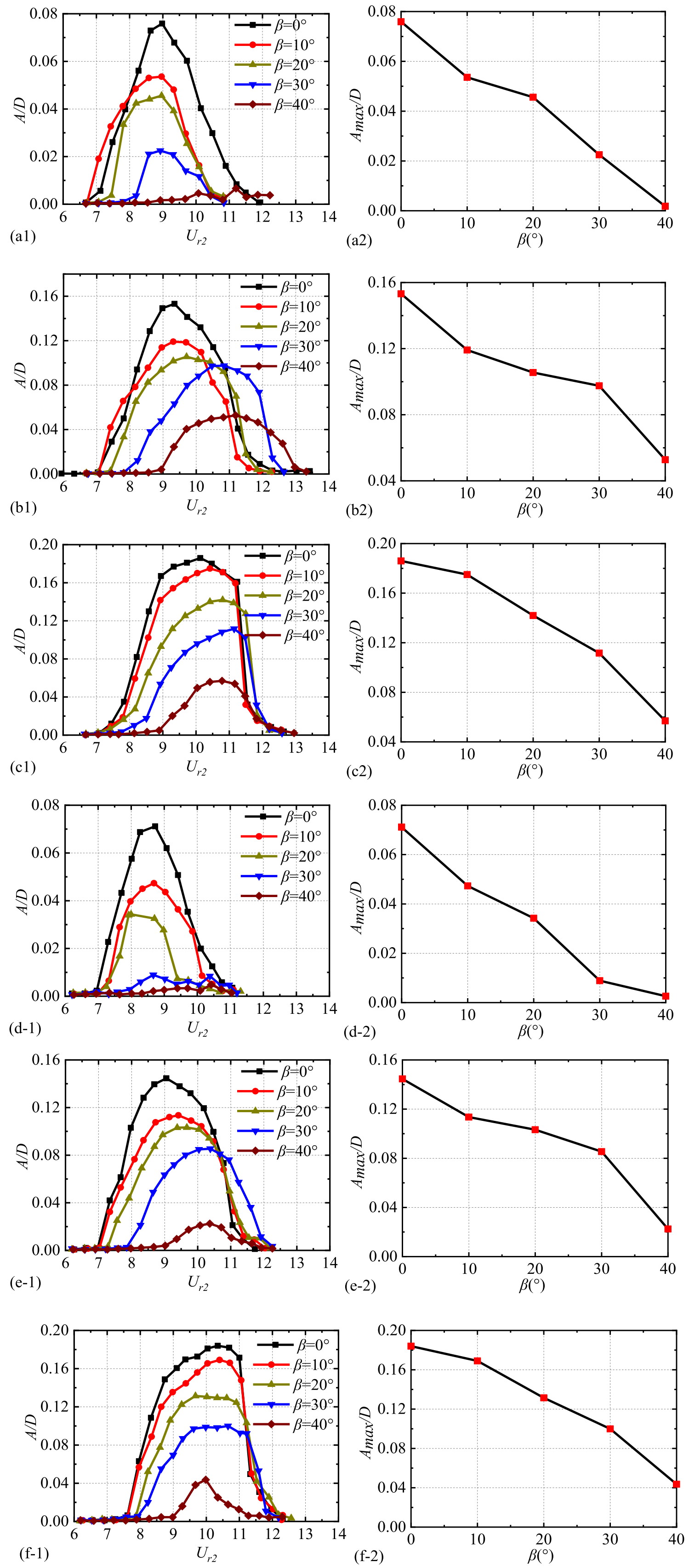

Fig. 15. Variations of VIV lock-in region and peak amplitude with wind yaw angle: (a1) - (a2) α_m = -5° under smooth flow; (b1) - (b2) α_m = 0° under smooth flow; (c1) - (c2) α_m = +5° under smooth flow; (d1) - (d2) α_m = -5° under turbulent flow; (e1) - (e2) α_m = 0° under turbulent flow; (f1) - (f2) α_m = +5° under turbulent flow.

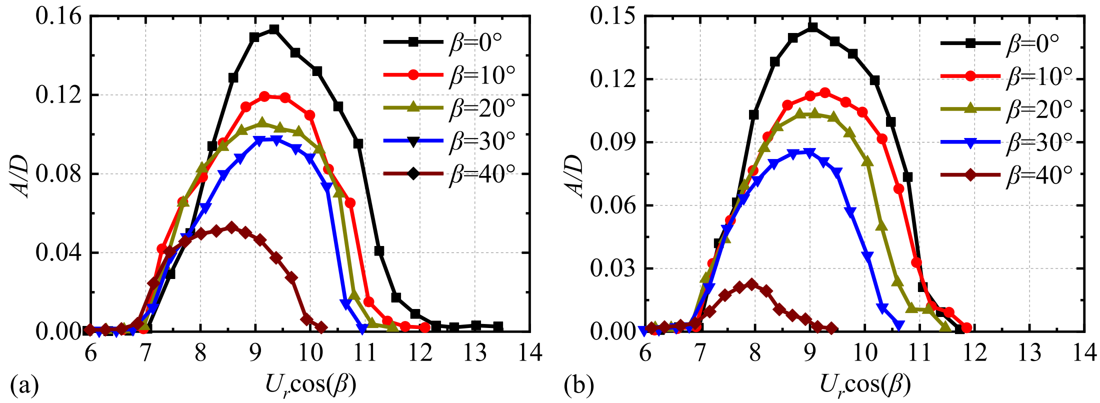

Fig. 16. The curves of VIV dimensionless amplitudes versus $U_r\cos\beta$ when $\alpha_m = 0°$:(a) smooth flow; (b) turbulent flow.

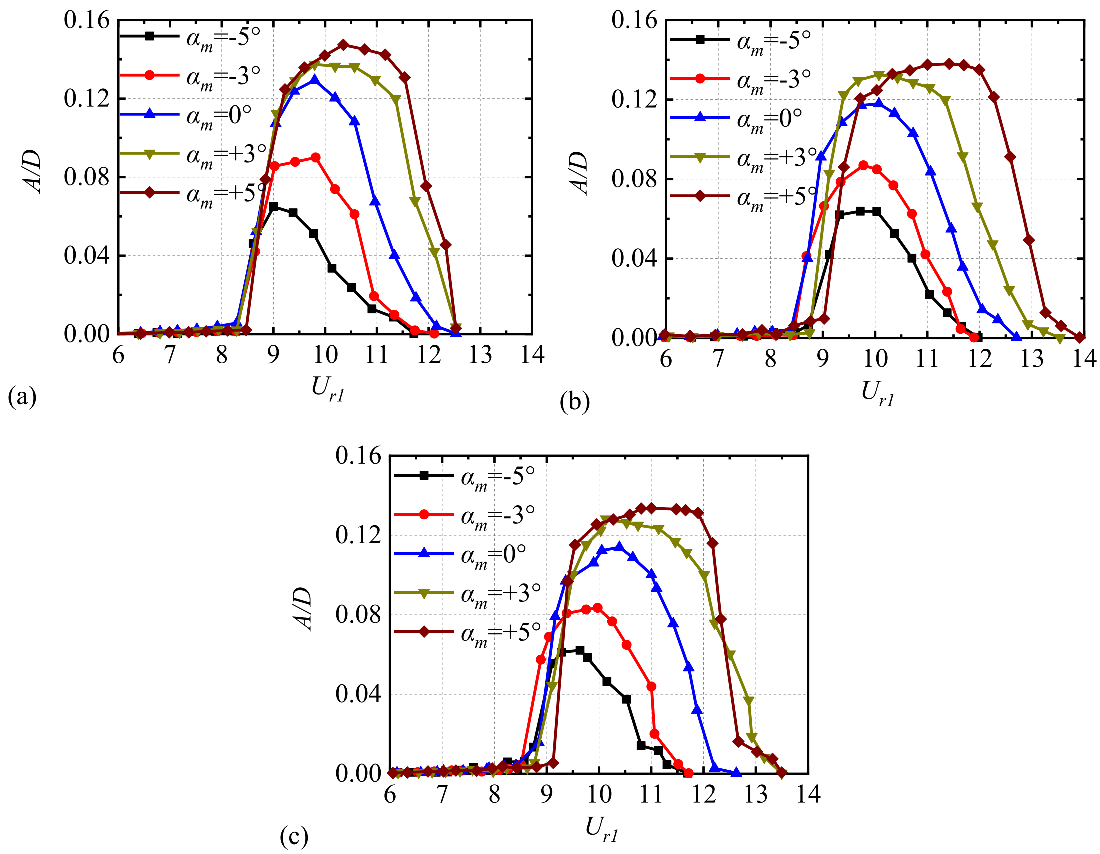

Fig. 17. The VIV responses of the sectional model: (a) smooth flow; (b) I_u= 4.27 %; (c) I_u= 5.67 %.

the VIV performances of the full-bridge aeroelastic model were analyzed by the conditions of smooth flow and turbulent flow when α_m = -5°, 0° and + 5°. Fig. 15 shows the A/D and the peak amplitudes (A_{max}/D) vary with β for various cases.

The variations of VIV response under smooth flow are summarized as follows. When β grows from 10° to 20°, the A_{max}/D drops gradually for α_m = -5°. The lock-in region disappears when β = 40°. At $\alpha_m = 0°$, as β grows from 10° to 30°, the A_{max}/D decreases slowly. The onset and cut-off wind speed are significantly delayed when $\beta = 30°$ and 40°. When β grows from 0° to 10°, the A_{max}/D drops gradually for $\alpha_m = 5°$. The onset wind speed progressively increases while the cut-off wind speed remains constant. Comparable occurrences are witnessed in turbulent flow, although the findings are not elaborated upon for conciseness. It is important to mention that when α_m = -5°, the lock-in region ceases to exist when $\beta = 30°$. When $\alpha_m = 0°$ and $\beta = 40°$, the A_{max}/D decreases significantly, so the cut-off wind speed is different from that of the smooth flow. When $\alpha_m \neq 0°$, the actual wind angles of attack (α) also change with β. Therefore, the decreasing rates of the A_{max}/D with β and lock-in regions differ slightly for different α_m. Overall, under various α_m and flow conditions, as β increases, the VIV amplitude typically decreases, although the reduction rates may vary, and the onset wind speed for VIV shifts.

The actual wind angle of attack will change with β when $\alpha_m \neq 0°$, so the VIV responses under skew wind at $\alpha_m = 0°$ are utilized to confirm the applicability of the IP theory [22]. The VIV responses at different β are plotted versus $U_r\cos\beta$ when $\alpha_m = 0°$ under smooth flow and turbulent flow in Fig. 16.

The IP theory predicts that VIV responses should be roughly consistent at different β. As shown in Fig. 16, significant differences exist between the results at different β relating the lock-in region and the

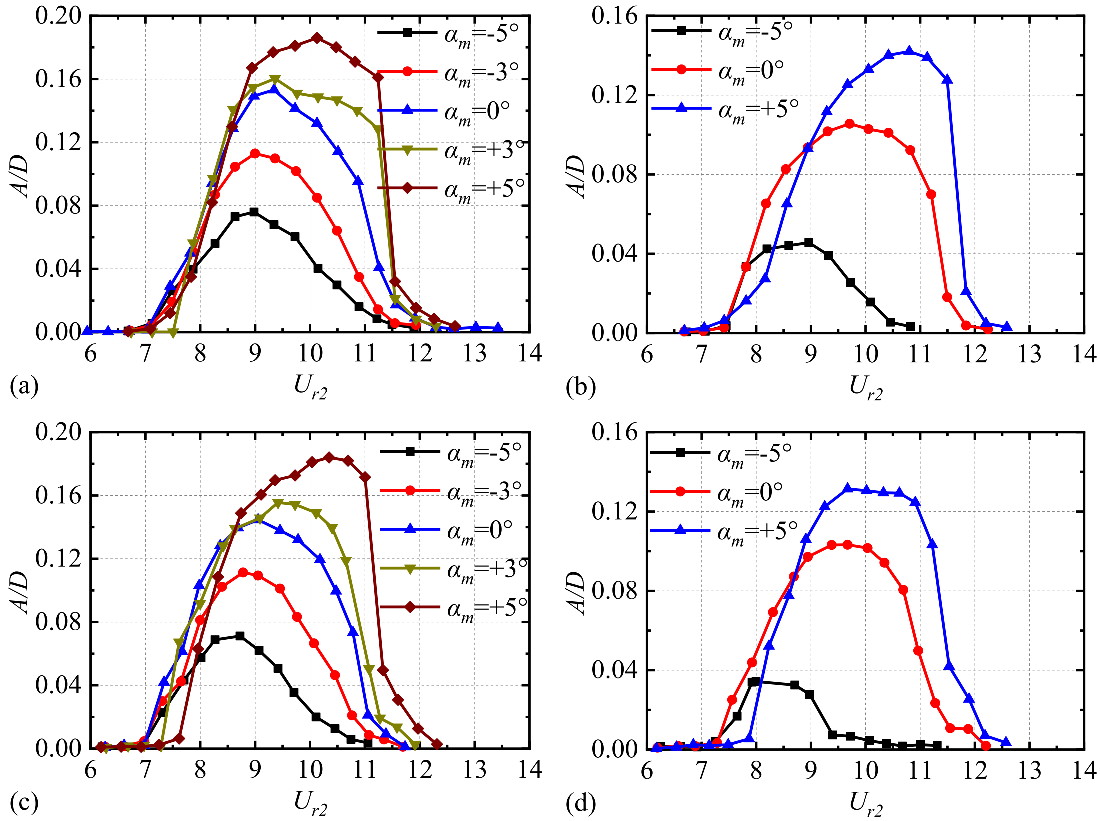

Fig. 18. The VIV responses of the full-bridge model: (a) smooth flow $\beta = 10^{\circ}$; (b) smooth flow $\beta = 20^{\circ}$; (c) turbulent flow $\beta = 10^{\circ}$; (d) turbulent flow $\beta = 20^{\circ}$.

maximum A/D. The onset wind speed of the lock-in region remains constant at varying β. Nevertheless, the cut-off wind speed experiences significant changes as β increases. As the β increases, the lock-in region shrinks, and the A_{max}/D decreases. When $\beta \neq 0$, the skew wind's perpendicular and parallel wind components interact and prevent regular vortex shedding. Therefore, the experiment's results indicate that the IP theory cannot be used to calculate the VIV response under skew winds of the studied bridge.

3.3. Influence of α_m on VIV

The influences of α_m on A/D of the sectional model and full-bridge models under different conditions are shown in Fig. 17 and Fig. 18, respectively.

With α_m changing from -5° to $+5^{\circ}$, the lock-in region extends slightly for both the sectional and full-bridge aeroelastic models. The onset reduced velocity remains almost unchanged, but the cut-off reduced velocity increases with α_m changing from -5° to $+5^{\circ}$. The A_{max}/D increases, and the corresponding wind speed is delayed. The A/D decays faster after reaching the maximum amplitude for a larger α_m. The results shown in Fig. 17 and Fig. 18 suggest that the most unfavorable conditions for VIV occurrence in similar bridge decks may arise at positive wind angles of attack. Similar phenomena are observed in turbulent flow cases, although the results are not presented here for brevity. It is worth noting that the A_{max}/D at the amplitude-rising stage of small α_m may exceed the A_{max}/D of large α_m in turbulent flows. All in all, with the increase of α_m, the A_{max}/D decreases, and the VIV lock-in region narrows.

3.4. Influence of turbulence intensity on VIV

The influences of Iu on A/D of the sectional model were analyzed based on three sets of testing cases (i.e., set with $\alpha_m = -5^{\circ}, -3^{\circ}, 0^{\circ}, +3^{\circ}$, and $+5^{\circ}$, respectively), as shown in Fig. 19. The lock-in region, A_{max}/D and corresponding wind speed of each testing case are listed in Table 5.

From Table 5, the A_{max}/D decreases with the increase in turbulence intensity, but the decrease is not significant. The A_{max}/D corresponding wind speed increases with the turbulence intensity. The VIV lock-in region has no apparent alteration as the turbulence intensity increases. Comparable phenomena are witnessed for different α_m, although the results are not included for brevity.

The influences of Iu on A/D of the full-bridge aeroelastic model were analyzed based on testing cases with different combinations of α_m and β, as shown in Fig. 20.

As shown in Fig. 20(a), when $\beta = 0^{\circ}$, the A_{max}/D in the turbulence flow at different α_m is smaller than that in the smooth flow, but the reduction is insignificant. The lock-in region has no noticeable change with the increase of the turbulence intensity. In Fig. 20(c), the VIV characteristics under different flow fields when $\beta = 20^{\circ}$ are the same as those at $\beta = 0^{\circ}$, but the A_{max}/D reduction is more evident at $\alpha_m =$ -5° and + 5°, and the lock-in region range in turbulence is remarkably narrowed when $\alpha_m =$ -5°. The turbulence intensity at each point of the full-bridge aeroelastic model varies due to the presence of β during the turbulent condition test. Changing β also alters the aerodynamic shape of the model, and since the VIV response is highly sensitive to the details of the cross-section, the VIV characteristics between $\beta = 40^{\circ}$ to 0° are slightly different. In general, turbulence intensity has little effect on the VIV characteristics for this bluff body section. Therefore, the suppression effect of on-site turbulence on the VIV amplitude cannot be overestimated for the blunter section. Comparable phenomena are witnessed for different β, although the results are not included for brevity.

3.5. Influence of Sc number on VIV

An analysis was conducted on the effects of the Sc number on the VIV performance of the sectional model under smooth flow conditions with

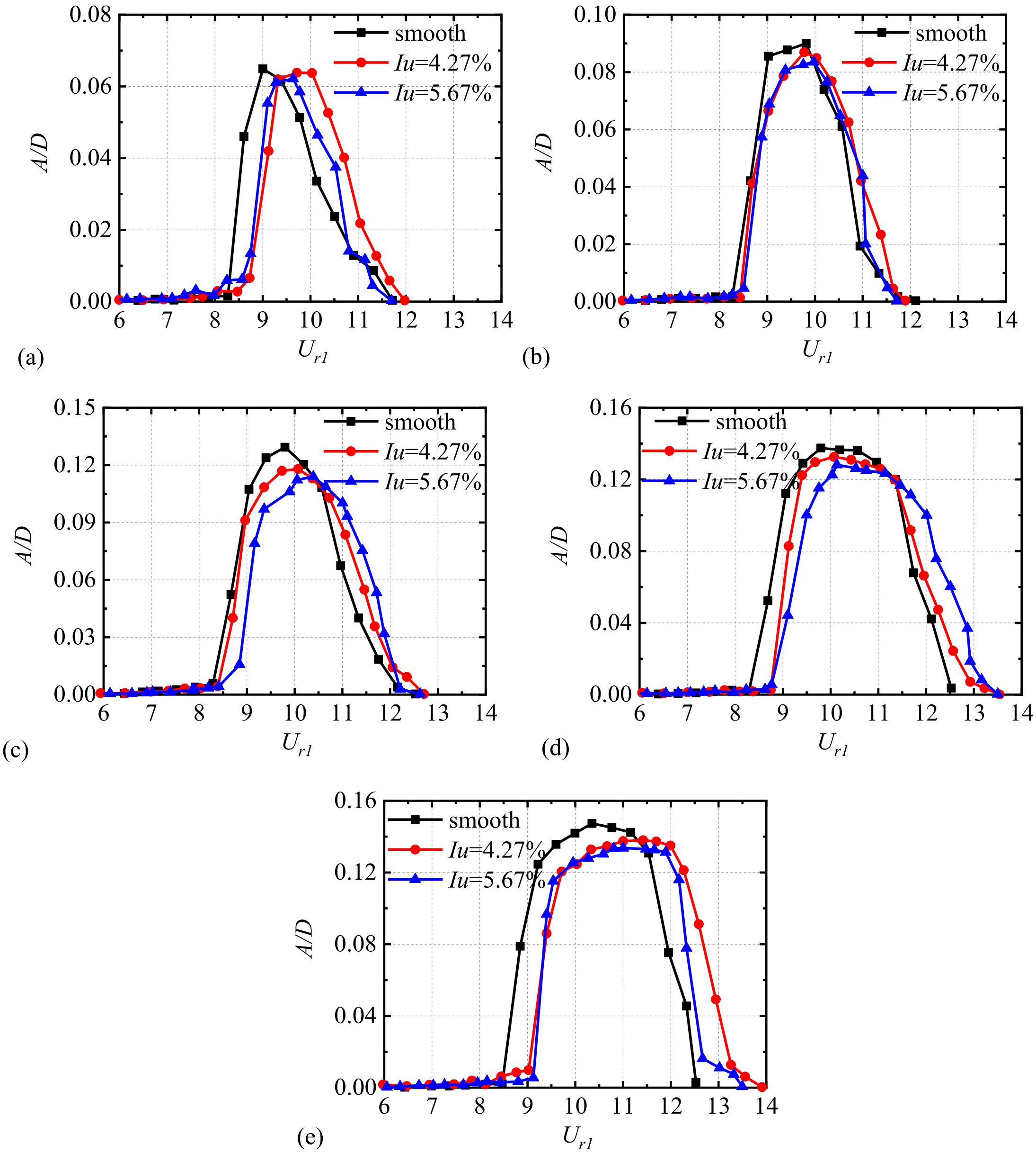

Fig. 19. The VIV responses of sectional models when Sc = 0.528: (a) α_m = -5°; (b) α_m = -3°; (c) α_m = 0°; (d) α_m = +3°; (e) α_m = +5°.

Table 5
Test results of all cases.

α_m	Iu	lock in region	A_{max}/D	corresponding U_{r1}
−5°	smooth	(8.27,11.71)	0.065	9.00
	Iu= 4.27 %	(8.47,11.97)	0.063	9.71
	Iu= 5.67 %	(8.27,11.71)	0.062	9.63
−3°	smooth	(8.27,12.11)	0.090	9.81
	Iu= 4.27 %	(8.43,11.89)	0.087	9.77
	Iu= 5.67 %	(8.27,11.71)	0.083	9.97
0°	smooth	(8.29,12.62)	0.129	9.79
	Iu= 4.27 %	(8.41,12.70)	0.117	10.07
	Iu= 5.67 %	(8.41,12.62)	0.113	10.39
+ 3°	smooth	(8.23,12.52)	0.137	9.79
	Iu= 4.27 %	(8.74,13.53)	0.132	10.07
	Iu= 5.67 %	(8.76,13.49)	0.128	10.13
+ 5°	smooth	(8.45,12.52)	0.147	10.35
	Iu= 4.27 %	(8.45,13.91)	0.138	11.41
	Iu= 5.67 %	(9.12,13.49)	0.133	11.00

α_m = 0°. The Sc number is changed by adding additional viscous damping to the sectional model. Fig. 21 (a) shows the dimensionless amplitudes of the sectional model under different Sc numbers. When considering different Sc numbers, the lock-in region's cut-off wind speed is approximately U_{r1} = 11.73. For $Sc \leq 16.63$, the onset wind speed is about U_{r1} = 8.27. However, for Sc > 16.63, the onset wind speed will increase as the Sc numbers grow. At the lower Sc numbers ($Sc \leq 11.72$), there is a sudden decrease in amplitude right after the peak amplitude is reached. By contrast, for the higher Sc numbers ($Sc > 11.72$), a gradual reduction of the response amplitude is observed with a more asymmetric response pattern around the peak. In Fig. 21 (b), the A_{max}/D falls as the Sc number increases. Specifically, when the Sc increases from 0.54 to 7.41, A_{max}/D decreases by 28 %, and when Sc increases to 32.84, A_{max}/D decreases by 73 %. All in all, the maximum A_{max}/D decreases as the Sc number increases. The variation curve of the A_{max}/D with the Sc number aligns with the results for a 4:1 rectangular cylinder, as reported by Marra et al. [27]. The A_{max}/D decreases as the Sc number increases, although the reduction rate slows down. This difference in reduction rates is attributed to the sensitivity of the aerodynamic configuration to

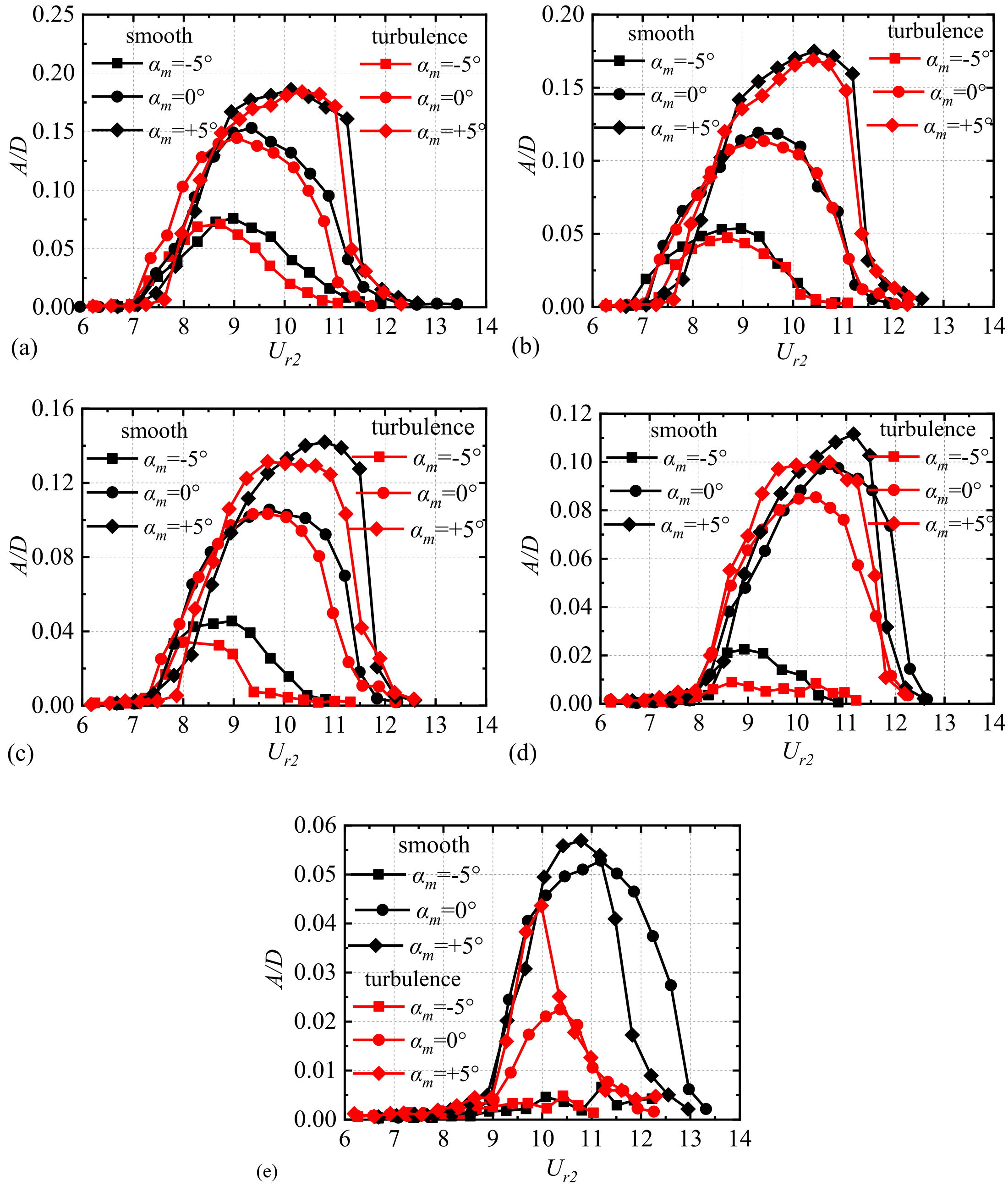

Fig. 20. The VIV responses of full-bridge aeroelastic models:(a) $\beta = 0°$; (b) $\beta = 10°$; (c) $\beta = 20°$; (d) $\beta = 30°$; (e) $\beta = 40°$.

the small size of the sectional model cross-section.

The influences of *Sc* numbers on *A/D* of the full-bridge aeroelastic model were analyzed by the condition of smooth flow when $\beta = 0°$. Fig. 22 shows the dimensionless amplitudes of the full-bridge aeroelastic model under different damping ratios.

Comparing case group FA with case group FC, when the *Sc* numbers grow from 0.604 to 2.104, the lock-in regions at each α_m are unchanged. The A_{max}/D decreases, and the lock-in region for large *Sc* number is enveloped in small *Sc* number. When the *Sc* numbers continuously increase to 3.344, VIV disappears. In summary, the A_{max}/D diminishes as the *Sc* number grows. Additionally, the VIV will cease when the *Sc* number reaches a crucial value. The VIV response is highly sensitive to the specific details of the section, meaning that the results for the studied section may differ from those of other sections.

4. Concluding remarks

This study presents a comprehensive analysis of vortex-induced vibrations (VIV) in a long-span continuous bluff steel box girder bridge, using 1:30 sectional and full-bridge aeroelastic models. The main findings and contributions from this study are as follows:

(1) As β grows, the peak amplitude (A_{max}/D) and the onset wind speed decrease and increase, respectively. The orthogonal wind condition produces the largest VIV response, representing the worst-case scenario. Under skew wind conditions, the traditional independence principle (IP theory) fails to predict the VIV response of this bridge type, emphasizing the need for alternative predictive methods for complex wind scenarios.

(2) The onset wind speed for VIV remains nearly independent of α_m, while the cut-off wind speed increases with higher α_m.

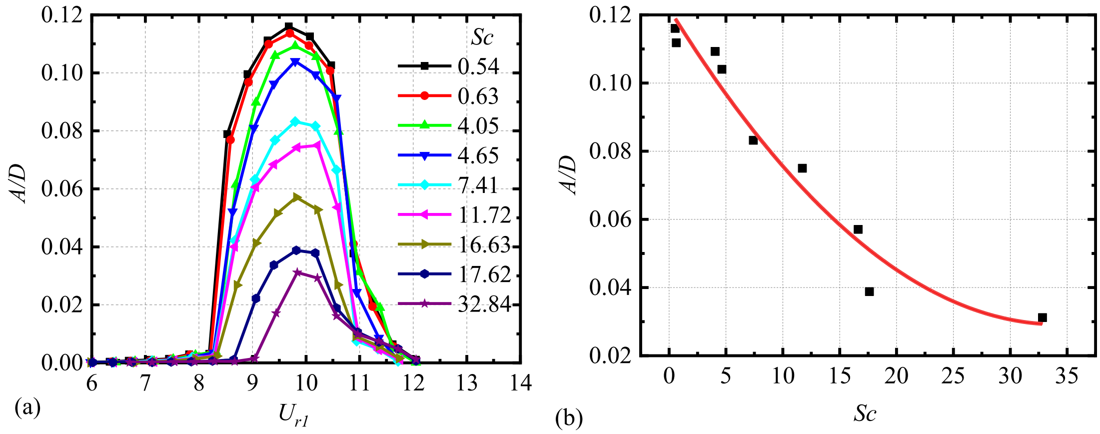

Fig. 21. The VIV responses of sectional model under smooth flow when $\alpha_m = 0°$: (a) Dimensionless amplitudes; (b) The A_{max}/D versus Sc.

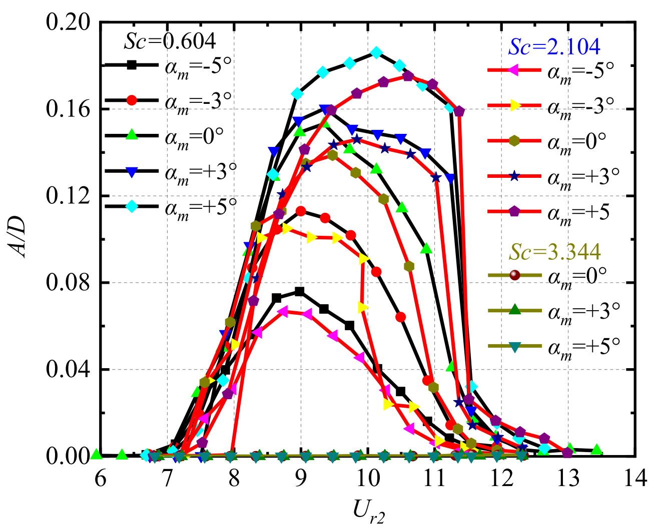

Fig. 22. The VIV responses of full-bridge aeroelastic models under smooth flow when $\beta = 0°$.

Additionally, low turbulence intensity ($Iu < 8$ %) has an insignificant effect on the concerned deck's VIV amplitude, validating that turbulence has limited damping effects on VIV, challenging conventional assumptions about turbulence mitigation.

(3) The A_{max}/D decreases with the damping ratio (or Sc number) increase. For the sectional mode test, at the lower Sc numbers ($Sc \leq 11.72$), a sharp fall of the amplitude is observed immediately after the peak amplitude is reached. By contrast, for the higher Sc numbers ($Sc > 11.72$), a gradual decrease in the response amplitude is observed with a more asymmetric response pattern around the peak. When Sc increases from 0.53 to 32.84, A_{max}/D decreases by 73 %, providing a quantitative benchmark for enhancing bridge resilience. For the full-bridge model, when Sc increase to 3.344, VIV is completely eliminated, establishing a critical threshold for practical design considerations to mitigate VIV.

CRediT authorship contribution statement

Fuyou Xu: Writing – original draft, Validation, Supervision, Software, Methodology, Conceptualization. **Zhanbiao Zhang:** Writing – review & editing, Writing – original draft, Methodology. **Bo Wang:** Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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